

# Ahmed Al-Hajri

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## Education

**University of Manchester, United Kingdom**

Sep 2023 – Jun 2026

*BSc (Hons) Computer Science and Mathematics*

**First-Class Honours (GPA: 4.0/4.0)**

Relevant Modules: Artificial Intelligence, Machine Learning, Algorithms & Data Structures, Software Engineering, Probability & Statistics, Algebra, Information Theory, Quantum Computing.

**Third-Year Dissertation: Towards Random Polynomial Systems through Iterated Composition of Gröbner Bases**

- Designed the *Iterated Gröbner Basis Composition with Shuffle* algorithm to generate high-complexity polynomial systems with controlled structural properties.
- Developed a machine learning classifier using 12 statistical features to distinguish structured and random polynomial systems across more than 20,000 generated instances.
- Combined computational algebra, algorithm design, and machine learning to investigate the structural evolution of polynomial systems under iterative composition, tackling an open problem in computational algebra.

**INTO Manchester (NCUK Foundation)**

Sep 2022 – Jun 2023

International Foundation in Mathematics and Computer Science

Grades: **A\*A\*A\***

## Technical Skills

**Programming & Data:** Python, SQL, MATLAB, R, Machine Learning, Predictive Modelling, Time-Series Analysis, Statistical Analysis, Data Visualisation

**Engineering & DevOps:** Artificial Lift Data Analysis, Production Data Analytics, Optimisation, Git, Docker.

## Experience

**Artificial Intelligence Intern, Occidental Oman (Oxy)**

Jun 2026 – Sep 2026

- Developed AI and machine learning solutions addressing three real-world production engineering challenges within the oil and gas industry.
- Built predictive models using operational and production data, collaborating with production and petroleum engineers to support data-driven decision making.
- Presented technical findings to engineers and stakeholders, translating machine learning results into practical recommendations for field operations.

**Teaching Assistant – Programming & Machine Learning, University of Manchester**

Oct 2025 – May 2026

- Facilitated learning for more than 500 first- and second-year students across programming and machine learning laboratories.
- Guided students through debugging, algorithmic problem-solving, supervised learning concepts, and model evaluation techniques, enhancing their programming and analytical skills.
- Collaborated with academic staff to develop and implement consistent teaching practices and assessment standards.

**Research Intern, Project X Agency, University of Manchester**

Jun 2025 – Sep 2025

- Selected for the competitive Learning Through Research Student Experience Internship within the Faculty of Science and Engineering.
- Co-designed a university module preparing students with digital skills for Industry 5.0 through literature reviews, focus groups, and practical workshops.
- Facilitated the Emirates Airlines sustainable cabin renewal case study, connecting academic research with industrial challenges.

## Leadership Experience

**Logistics Officer, University of Manchester Mathematics Society**

Sep 2024 – Jul 2025

- Coordinated logistics for academic and social events serving more than 900 members.
- Redesigned the society website to improve accessibility and student engagement.

**President & Vice President, Omani Students Society in Manchester**

Aug 2022 – Dec 2023

- Directed over 20 cultural, academic, and professional events attended by more than 1,000 participants.
- Secured sponsorships and led the society's growth into one of the UK's largest Gulf student societies.

## Awards

- National Technology Competition Finalist (Top 200 nationwide for an award-winning software application).
- Winner, Cultural Creativity Competition (Youth Innovations).